Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory were checked in Lean, where the formal version differs from the brief, and any problems found in the brief. The brief is available at `https://assumptionsofphysics.org/resources/briefs`.

Setup

- Lean `v4.36.0-rc1`; Mathlib at commit `2cabef97`; physlib (`master`) and TauCeti (commit `eb4dd24`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Unphysicality of Hilbert spaces for quantum mechanics

Files.

- File: `briefs/HilbertUnphysicality/HilbertUnphysicality.lean`
- Library: `HilbertUnphysicality`
- Imports: `InconsistentConvergence` and three new `ForMathlib` files:
  `ForMathlib/Analysis/SpecialFunctions/Gaussian/Erf.lean` (the error function),
  `ForMathlib/Analysis/InnerProductSpace/l2Space.lean` (separable Hilbert spaces are $\ell^2(\mathbb{N})$),
  `ForMathlib/MeasureTheory/Function/L2Space.lean` (L^2 is infinite-dimensional)

Setting. Theorem 5 says that the Hilbert-space formulation QSTO:H is inconsistent with three physical conditions. The brief proves four mathematical propositions (Sec. 2) and then argues each part of Theorem 5 from one of them (Sec. 4). The propositions are checked in Lean; Theorem 5 is checked by reading, against them.

Results.

Brief	Lean
Proposition 1	
$\mathcal{D}(A) \neq \mathcal{H}$ (Hellinger–Toeplitz)	`bounded_of_isSymmetric`, `not_isSymmetric_of_unbounded`
A is not continuous on $\mathcal{D}(A)$	`not_continuous_of_unbounded`
Proposition 2	
$x_j \to e_0$, $E_N(x_j) = 1$, $E_N(e_0) = 0$	`numberExp_not_continuousAt` (from the inconsistent-convergence brief)
Proposition 3	
erf, $\lvert \operatorname{erf} \rvert < 1$, $\operatorname{erf} \to \pm 1$	`Real.erf`, `Real.abs_erf_lt_one`, `Real.tendsto_erf_atTop`
$f(x) = \tan(\frac{\pi}{2} \operatorname{erf} x)$ is increasing	`frameChange`, `strictMono_frameChange`
$U\psi = \phi$, at the level of densities	`cauchyDensity_frameChange_mul_deriv`: $\lvert\phi(f(x))\rvert^2 f'(x) = \lvert\psi(x)\rvert^2$
the same, integrated	`arctan_frameChange`: $\arctan(f(x))/\pi = \operatorname{erf}(x)/2$
ψ normalized, $\psi \in \mathcal{D}(X)$	`integral_gaussDensity`, `integrable_sq_mul_gaussDensity`
ϕ normalized, $\phi \notin \mathcal{D}(Y)$	`integral_cauchyDensity`, `not_integrable_sq_mul_cauchyDensity`
Proposition 4	
separable infinite-dimensional Hilbert spaces are isomorphic	`nonempty_linearIsometryEquiv_of_separable`
$L^2(\mathbb{R}^n)$ is separable and infinite-dimensional	`L2.separableSpace_euclidean`, `L2.not_finiteDimensional_euclidean`
$L^2(\mathbb{R}^{n+1}) \cong L^2(\mathbb{R}^{m+1})$	`L2_euclidean_iso`

Differences from the brief.

- *Proposition 1.* Both parts are stated for linear maps, without Mathlib's partially defined operators. Discontinuity is proved for an unbounded linear map on any normed space, which includes $\mathcal{D}(A)$ with the norm inherited from $\mathcal{H}$. Hellinger–Toeplitz is proved for an everywhere-defined symmetric map; self-adjointness is not needed.

- *Proposition 2.* The number operator is not formalized as an unbounded self-adjoint operator. Its expectation is `numberExp` $x = \sum_n n\,|x_n|^2$ on all of ℓ^2 (the sum is 0 where it diverges), which agrees with E_N on normalized vectors of $\mathcal{D}(N)$. The sequence x_j is the brief's.

- *Proposition 3.* The wave functions are handled through their densities $|\psi|^2 = e^{-x^2}/\sqrt{\pi}$ and $|\phi|^2 = 1/(\pi(1+y^2))$. "In the domain of the position operator" is read as $\int x^2|\psi|^2 < \infty$, i.e. $x\psi \in L^2$. That f is strictly increasing is proved; that it is smooth and onto $\mathbb{R}$ is not. The unitary U, its inverse V, the change of variables, and the domain of UXU^{-1} are not formalized. Mathlib has no error function; `Real.erf` is defined in `ForMathlib`.

- *Proposition 4.* The classification of separable infinite-dimensional Hilbert spaces is new in `ForMathlib`. The positive integers n, m appear as $n+1, m+1$.

- *Theorem 5.* Not stated in Lean; checked by reading. Each part is a direct application of a proposition, for one particular system: the oscillator (Proposition 2), one spatial degree

of freedom (Proposition 3), and two systems with different numbers of continuous degrees of freedom (Proposition 4). That a unitary carries rays, density operators, self-adjoint operators, the Born rule and the entropy is not formalized. Neither are the closing remarks on classical cutoffs and on Schwartz spaces.

- The additional remarks of Section 5 are not formalized.

Problems found in the brief.

- Cross-references lack their label: the proof of Theorem 5 has "This satisfies the conditions for 2", "As shown in 3" and "By 4" for Propositions 2, 3 and 4.
- "Schwartz spaces present the same problem" is asserted without proof, and without saying which structure on $\mathcal{S}(\mathbb{R}^n)$ is meant (see the open problems).
- QSTO:H still allows any separable Hilbert space, finite-dimensional ones included. This no longer affects Theorem 5, whose proof reads the inconsistency as the failure of a condition for one particular system, but the brief could say so.